

Exercise 7

Computer Vision, Convolution

1. Convolution, Cross-Correlation

Consider the following 2D array A and the filter kernels h_1 and h_2

A

0	2	3	0	3
4	0	8	3	0
0	7	5	0	3
8	0	5	6	2
0	0	4	2	3

 h_1

2	1	-1	-2
2	1	-1	-2
2	1	-1	-2
2	1	-1	-2

 h_2

0	-1	0
-1	4	-1
0	-1	0

- Calculate the **2D convolution** of A with h_1 , using **'valid' mode**.
- Calculate the **cross-correlation** of A with h_2 , using **stride = 2** and **zero padding**.
- Interpret your results! What are the output dimensions of $A * h_1$ and $A \times h_2$ respectively?

2. Programming Task – 2D Convolution in C

Write a C program that performs a **2D convolution** operation on a matrix with an arbitrary filter kernel.

- Download and open the file `myConvolution.c`.
- The file contains predefined input matrices and function prototypes.
- Implement the missing parts** of the code and **compile the program** using the GNU C Compiler (`gcc`).

After implementing the convolution, **verify your results** by comparing them with your calculations from **Exercise 1a**.

Extension:

Enhance your program to support **configurable padding** and **stride values**.

3. Jupyter Notebook – Computer Vision

Continue working with Jupyter Notebooks.

Please open the notebook `CV_Exercise.ipynb` and follow the instructions provided within.

Note:

Make sure to also download the image files (`parot.jpg`, `zebra-small.jpg`) and save it to the same directory as your notebook (your workspace), so it can be loaded correctly.